

1. Mosquito Creek Claim #1117244

1.1 Geology and Rock Types

Mosquito Creek lies immediately downstream of the Cornish Mountain and Island Mountain lode belts, in a drainage fed by Snowshoe Formation metasediments of the Cariboo Group. The Snowshoe Formation is described as a sequence of schistose greywacke, siltstone, phyllite and thin, impure limestone lenses that form the upper part of the Cariboo Group in this area (08089.pdf/ p. 15). These rocks host replacement-style pyrite gold mineralization and quartz veins at Cariboo Gold Quartz, Island Mountain and the Coronada prospect on Cornish Mountain.

Gold City Mining Corporation reports that combined lode production from the Cariboo Gold Quartz and Island Mountain mines totals approximately 1,200,000 ounces of gold, with more than 600,000 ounces of this production from Island Mountain alone between 1934 and 1987 (24723.pdf/ Page. 1–3, 5). This level of production demonstrates the strength of the lode system feeding Mosquito Creek.

1.2 Claim Gold Potential

Mosquito Creek receives sediment and gold from an area with multiple documented lode sources, including the Coronada and Martins Creek workings on Cornish Mountain and the Island Mountain mine. A seismic refraction report on the Willow Property notes that Mosquito Creek has produced on the order of 200,000 ounces of placer gold (25883.pdf/ p.)

1.3 Workings, Sampling and Geochemistry

Historic underground work at the Coronada prospect included a tunnel of approximately 400 metres driven to test quartz veins exposed high on Cornish Mountain and a second tunnel of about 100 metres in the Martins Creek area to intersect pyritic quartz veins (Yuma report, 08605.pdf/ Page. 4–6, 11–12). Surface trenches above the main tunnel expose zones of quartz veining over a significant strike length, providing evidence of vein hosted gold updrainage from the placer claim.

Soil geochemistry in the Yuma report records gold in soils up to several tens of parts per billion Au in the grid north of the Coronada tunnel and a silver anomaly correlating with a quartz vein sample collected by R.W. Phendler that assayed 0.004 oz/ton Au and 11.20 oz/ton Ag (08605.pdf/ p. 18). The Ace Group soil survey on the southern slopes of Cornish Mountain describes anomalous zones of gold and silver in B-horizon soils and concludes that the results strongly suggest a source of precious metals within the Ace claims (09348.pdf/ Page. 16–17).

More recent work on the Island Mountain property includes the 925meter Jukes audit driven in 1988 from the 1,220-metre elevation in the Mosquito Creek valley to the old 4000 Level of the Island Mountain mine. In the first two hundred meters of this audit, a sericite-silica-pyrite alteration zone returned gold enriched drift rounds up to 900 ppb Au; however

a 1995 Gold City diamond drill program of two NQ holes totaling about 390.3 meters did not find significant gold enrichment and did not identify bulk-tonnage lode potential (24723.pdf/ Page. 2–3, 5–7). Although this work is on lode ground, it further illustrates the style and strength of mineralization in the Mosquito Creek Island Mountain belt that supplies gold to the Mosquito Creek placer system.

2. Lowhee Cow Mountain Claim #1114560

2.1 Geology and Rock Types

The Lowhee / Cow Mountain claim is situated on Lowhee Creek within the core of the Barkerville Gold Camp. Bedrock consists of Cariboo Group metasediments, including units equivalent to the Snowshoe Formation and Rainbow member that host lode deposits on Cow and Island Mountains. In the Wells area, lode gold is hosted predominantly by carbonate and metabasalt bearing rocks of the middle Paleozoic Downey Succession, one of several carbonate rich successions in the Barkerville Terrane (25883.pdf/ p. 3).

The same report notes that about 1.2 million ounces of lode gold has been mined from Cow Mountain and Island Mountain approximately 2KM east of the Willow Property, mainly from replacement pyritic carbonates and quartz veins (25883.pdf/ p. 3). These lodes provide a strong bedrock source for gold entering Lowhee Creek.

2.2 Claim Gold Potential

Lowhee Creek is widely recognized in regional placer summaries as one of the richest creeks in the Barkerville district. Long term records indicate placer production in the tens of thousands of ounces and gold recovered from the creek is described as coarse and relatively angular to sub rounded, consistent with short transport distances from nearby lode sources on Cow and Island Mountains.

Although no recent claim specific sampling or soil geochemical survey was included for this tenure in the supplied Ace and Yuma reports the combination of a 1.2-million-ounce lode camp immediately updrainage and a proven high grade placer channel supports a very high placer potential rating for Claim #1114560.

2.3 Workings, Sampling and Geochemistry

Historic mining on Lowhee Creek progressed from early hand methods and sluicing to more mechanized hydraulic and drift operations, recovering substantial gold from different segments of the channel. The coarseness and morphology of the gold, as described in regional placer reports, indicate a proximal bedrock source rather than long-distance reworking from more distal deposits.

No specific soil geochemical or bulk-sampling results for this claim were present in the Ace or Yuma reports. Instead, the placer potential assessment relies on the well-established lode sources in the Downey Succession and the documented historic placer production from Lowhee Creek.

3. Ballarat Claim #1112033

3.1 Geology and Rock Types

The Ballarat placer claim is located within the Barkerville Gold Camp in a belt of Cariboo Group metasedimentary rocks similar to those underlying Williams, Lowhee and Mosquito Creeks. These rocks include folded metasediments and limestone lenses within the same structural framework that hosts lode deposits elsewhere in the camp.

Seismic refraction work carried out on the Willow Property along the Willow River, approximately 3 kilometres northwest of Wells, indicates a thin surficial layer (about 2 metres thick) overlying a 16 – 37m thick intermediate layer with seismic velocities of about 2,130–2,150 m/s, interpreted as dense saturated sands, gravels, cobbles and boulders or dense glacial till, above a competent phyllite and siltstone bedrock surface (25883.pdf/ Page. 5–7). The interpreted bedrock surface along one of the seismic lines suggests that the ancient Willow River occupied a broad buried channel beneath the modern valley floor (25883.pdf/ Page. 7–8). Although no such geophysical data are available for Ballarat, these results demonstrate that deep, potentially gold bearing paleochannels can be preserved beneath present drainages in the immediate Barkerville area.

3.2 Claim Gold Potential

The Barkerville camp as a whole has produced several million ounces of gold from lode and placer deposits. The Willow seismic report records that Mosquito Creek has produced approximately 200,000 ounces of placer gold and Hardscrabble Creek about 2,000 ounces. High-grade gravels in the Willow River channel are reported to exceed 1 ounce of gold per cubic metre at depths around 30m (25883.pdf/ Page. 4, 7). These examples confirm that thick, deeply buried gravels can host significant placer gold accumulations in the district.

No Ballarat specific assessment report or sampling data were included in the documents reviewed, so no direct grades or gravel thicknesses can be quoted for this claim. Nevertheless, its position within a well endowed camp and in a favourable stratigraphic and structural setting justifies a high, but inferred, placer potential rating that will require confirmation by targeted field work.